

The Three-Point Revolution: A Quantitative Analysis of the NBA's Strategic Transformation

Adam Rychtecky

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Abstract

This study quantitatively examines the NBA's transition toward three-point shooting using over 4.2 million shot attempts spanning 21 seasons from 2004 to 2024. Using distributed processing and statistical modeling techniques, the analysis addresses three core questions: whether the shift was mathematically justified, when it accelerated, and how it reshaped player roles. Zone efficiency results show a clear structural disadvantage for mid-range shots, which yield substantially fewer points per attempt than three-point shots due to similar field-goal percentages. A LASSO-regularized classification model further demonstrates that shot location and type are the strongest predictors of shot success, outweighing contextual game factors. Time-series analysis reveals a steady rise in three-point attempts, with a pronounced inflection point in the mid-2010s. Finally, clustering of player shot profiles identifies five distinct archetypes and highlights a major shift in player composition, with mid-range specialists sharply declining and versatile, perimeter-oriented scorers becoming dominant. Overall, the findings indicate that the three-point revolution was a mathematically driven and inevitable transformation that fundamentally redefined offensive strategy and player value in the modern NBA.

1 Introduction

The NBA is one of the foremost sports leagues in the USA generating billions in revenue and having been running for nearly a century. Any person who wants to call themselves a fan of basketball knows about the transition from shooting 2pt shots to 3pt shots during the 2010s era of basketball. The so called analytics movement stated the 2pt mid range shot was “inefficient” compared to its 3pt counterpart sparking a new offensive era. Was the vanishing of this shot type and even player type justified? Or were the analytics missing a nuance on a professional player could know?

This basketball era transition always interested me as it seemed almost late. How did players, coaches, and teams not realize this inefficiency all the way until the 2010s? Decades of basketball where the most inefficient shot type was the most popular did not add up. This led me to pursue the following questions: was the shift toward the three mathematically justified; when exactly did the league tip to accepting this; and what happened to the types of players who had to endure this transition.

I pulled from the official NBA dataset of 4.22 million shot records over 21 seasons (2004-2024), every shot tagged with location and zone. The scale and cleanliness of the dataset lets me measure efficiency by zone over the entire revolution. There has been research prior to mine showing the efficiency argument by the Warriors team who spearheaded the transition. I simply mean to quantify it at a large modern scale through various methods of analysis being a shot-make classifier season-level time series, and a clustering of player profiles.

My report follows a single story: the mid-range penalty was always present in the math, which Sections 4.1 (zone efficiency) and 4.3 (the shot-make model) establish; the Warriors provided the first clear, winning demonstration and the league subsequently tipped, which Section 4.2 (the era time series) dates; and the roles that define success reorganized around the three-point arc, which Section 4.4 (the clustering) documents. Taken together, these pieces answer the central questions of justification, timing, and consequence. The results point in a stark direction: the mid-range shot is consistently dominated, the tipping point arrives in the mid-2010s, and an entire archetype of player once central to offense nearly disappears from the modern game.

2 Data

2.1 Source and Coverage

The primary dataset is NBA shot-location records for all 21 seasons from 2003–04 through 2024–25. Referred to throughout by the calendar year the season ended, for example, “2004” for 2003–04. Data were sourced from the publicly available NBA_Shots_04_25 repository (Samangy, 2025) via github https://github.com/DomSamangy/NBA_Shots_04_25, which aggregates shot-level records from the official NBA Stats API (stats.nba.com). The raw dataset comprises 4,231,262 shot attempts across 26 columns, including court coordinates (LOC_X, LOC_Y), shot distance, zone classifi-

cation (`BASIC_ZONE`), action type, shot type (2PT or 3PT), game context fields (quarter, minutes and seconds remaining), player and team identifiers, and a make/miss indicator.

Raw CSV files were uploaded to Google Cloud Storage (bucket: `pstat135-adam`) and ingested using Apache Spark on a Google Cloud Dataproc cluster. Spark’s schema inference confirmed the 26-column structure and zero null values across all fields providing a thorough dataset for analysis.

2.2 Cleaning and Feature Engineering

For data preparation, I removed 8,820 backcourt shots as haily marry attempts represented as, `ZONE_RANGE = “Back Court Shot”`, do not represent genuine shooting decisions. This yielded **4,222,442 analysis-ready records**. Seven feature columns were added: `SECS_REMAINING` (clock context as a single integer), `IS_3PT` and `SHOT_VALUE` (two/three-point indicators), `IS_CLUTCH` (fourth quarter or overtime with 120 seconds remaining), `SHOT_MADE_INT` (integer cast of the boolean target, required for Spark MLlib), `GAME_DATE_PARSED`, and `EXPECTED_PTS` (`SHOT_VALUE × SHOT_MADE_INT`). The `ACTION_TYPE` field, which carries dozens of raw values, was consolidated into five broad buckets (Dunk, Layup, Hook, Jump Shot, Other) via keyword matching to prevent sparse one-hot expansion in model training. The residual “Other” category accounts for approximately 1.4% of shots.

The cleaned dataset was written to GCS as Parquet, partitioned by season (`SEASON_1`), enabling easy season-level filtering in downstream analyses.

2.3 Distributed Processing Justification

A single season’s records fit comfortably on a laptop. My justification for Spark lies in the scope of the analyses performed here: zone efficiency aggregations over all 4.2 million shots, a 21-season time-series, a binary classifier trained on the full record pool, and PCA plus K-Means over all qualifying player-seasons. Spark’s lazy evaluation model creates a logical plan as discussed in class and executed in a single distributed pass, which is the appropriate tool for cross-era analyses that must treat all seasons as one coherent dataset.

2.4 Known Limitations

Three data limitations are relevant to interpretation. First, the dataset contains no defender or contest information. This means the shot-make model measures the value of shot *location and action type*, not the quality of shot selection under defensive pressure. Second, the 2012 season was shortened by a labor lockout to 66 games (versus the standard 82), and the 2020 season was shortened and played in a bubble environment; both seasons show lower total attempt counts in the underlying season-by-season data and should not be interpreted as trend deviations. Third, cross-era rule changes, most notably the 2004–05 defensive hand-check rule tightening, may affect FG% comparisons between early and late seasons in this panel due to changes in defensive level.

3 Methodology

3.1 Zone Efficiency

Zone efficiency was computed by grouping all shots by `BASIC_ZONE` and calculating attempt count, field-goal percentage (FG%), and points per attempt ($\text{pts/attempt} = \text{FG\%} \times \text{SHOT_VALUE}$). Points per attempt is the theoretically correct efficiency metric because it accounts for the value difference between two- and three-point shots, enabling a fair comparison: a 33.3% three-point shooter and a 50% two-point shooter are equally efficient at 1.0 pts/attempt. Six zones are present after backcourt removal: Restricted Area, In The Paint (Non-RA), Mid-Range, Left Corner 3, Right Corner 3, and Above the Break 3.

3.2 Era Analysis

Three-point attempt rate ($3\text{PT attempts} / \text{total attempts} \times 100$) was aggregated by season across all 21 seasons. The resulting time series (Figure 1) spans the pre-analytics era through peak three-point adoption, capturing the full arc from 2004 to 2024.

3.3 Shot-Make Model

I chose a binary classification model to predict whether a given shot is made. Three columns were excluded from the feature vector as leakage: `EXPECTED_PTS`, `EVENT_TYPE`, and `SHOT_MADE`. All of these provide explicit information on whether a shot is made hence their removal. The **feature vector** (26 dimensions) comprises eight numeric features: `LOC_X`, `LOC_Y`, `SHOT_DISTANCE`, `SECS_REMAINING`, `IS_3PT`, `SHOT_VALUE`, `IS_CLUTCH`, `QUARTER`, and four categorical features: `BASIC_ZONE`, `SHOT_TYPE`, `ZONE_RANGE`, and `ACTION_GROUP`. All transformers were fit on the 80% training split and applied to the held-out 20% test split (random seed 42).

LASSO regularization path. Feature importance was assessed adjusting the constricting L1 penalty over the following values [0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001], yielding respective feature counts of $3 \rightarrow 5 \rightarrow 9 \rightarrow 16 \rightarrow 21 \rightarrow 26$. The order in which features re-enter the model as regularization loosens shows the collinearity of certain features. Headline coefficients are reported at `regParam` = 0.001 (16 of 26 features active), chosen as the penalty that best preserves the zone structure without selecting all features back in.

Classifiers. `LogisticRegression` (`L2`, `regParam` = 0.01) and `LinearSVC` (`regParam` = 0.01) were trained on the same feature vector and evaluated on the held-out test split using ROC-AUC and macro-F1. All analyses used the full dataset encompassing all 4,222,442 observations.

3.4 Player Archetypes

Player archetypes were identified by clustering player-season shot profiles. Each player-season was represented as a six-dimensional vector of zone shot shares (rim, paint, mid-range, left corner 3,

right corner 3, above-the-break 3), which sum to one. The data was standardized, followed by PCA to reduce dimensionality, retaining 5 components. K-Means clustering was then applied with ‘k=5’ (random seed 42). Era stability was evaluated by holding cluster assignments fixed and comparing the distribution of player-seasons across archetypes before 2015 ($\text{SEASON_1} < 2015$) versus 2015 and later ($\text{SEASON_1} \geq 2015$).

4 Results

4.1 Zone Efficiency: The Mid-Range Penalty

Table 1 reports field-goal percentage and points per attempt for each of the six shot zones across all 4,222,442 shots in the dataset. Notably the Restricted Area is the most efficient with a Pts/Attempt of 1.232 which makes practical sense. The table also shows the inefficiency of the mid-range being the lowest Pts/Attempt at 0.798 a solid margin away from the lowest 3pt range being Above the Break at 1.054. This table perfectly shows in numeric form why the 3pt takeover made statistical sense for the league. With both shot types being near same percentage (39.9% vs 35.1%), the extra 1pt from the three is obviously worth the slightly lower make percentage.

Table 1: Zone efficiency across 4,222,442 shots (2004–2024). Points per attempt is field-goal percentage multiplied by shot value (2 or 3), the appropriate cross-zone efficiency metric. Zones are sorted by descending efficiency.

Zone	Attempts	FG%	Pts/Attempt
Restricted Area	1,344,005	61.6%	1.232
Right Corner 3	146,724	38.8%	1.164
Left Corner 3	159,656	38.6%	1.159
Above the Break 3	890,037	35.1%	1.054
In The Paint (Non-RA)	644,179	40.9%	0.818
Mid-Range	1,037,841	39.9%	0.798

4.2 Era Inflection: When the League Changed

Figure 1 plots the three-point attempt rate by season as a time series. Total shot attempts in the league remained stable around 200,000 each year. Looking at share of field goal attempts that are 3pt shots reflects era change from the mid range to the 3pt shot. The figure shows roughly flat around 22% up until 2012 where it begins to steadily increase with the sharpest increase being from 2015-2017. This time is also the Warriors Dynasty where their star player Stephen Curry became one of the first players to dominate strictly from the 3pt line. This team along with the stats shown in the figure represent the physical evidence of the league changing its typical play style to reflect this more efficient game plan. Additionally to be noted, the increase in 3pt shot take percentage has

yet to fully plateau reaching a high of 39.4% in 2024. Whether the league has found a ceiling is yet to be seen but due to defense adjustment along with other factors is likely to flatten eventually.

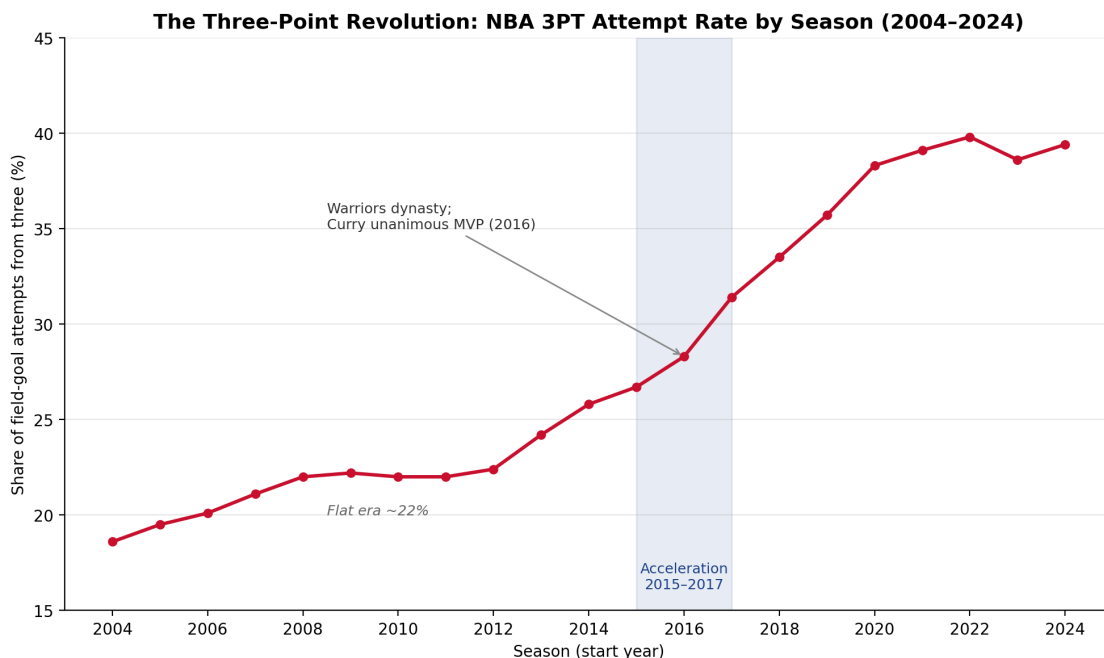


Figure 1: Three-point attempt rate by NBA season (three-point attempts as a share of total field-goal attempts), computed over all 4.22 million shots in the panel. Seasons are labeled by the year they ended; 2004 denotes the 2003–04 season. The shaded band and the 2016 annotation are added to mark the acceleration window and a contextual milestone, and are not features of the data. Because the metric is a normalized rate, the lockout-shortened 2012 and bubble-shortened 2020 seasons are unaffected by their lower game counts.

4.3 Shot-Make Model: Location Over Context

Figure 2 reports the headline LASSO coefficients at `regParam = 0.001`. Most notably and supporting the current shot make model action and location type features are the only prevalent ones with only one contextual feature making it (Clutch Situation). In addition to the figure, when the parameter is at its strongest only 3 features survive none of them being contextual. This in turn shows that the location and type of shot are the best predictors of a shot make further supporting the statistical idea of moving the area a shot is made to increase point acquisition.

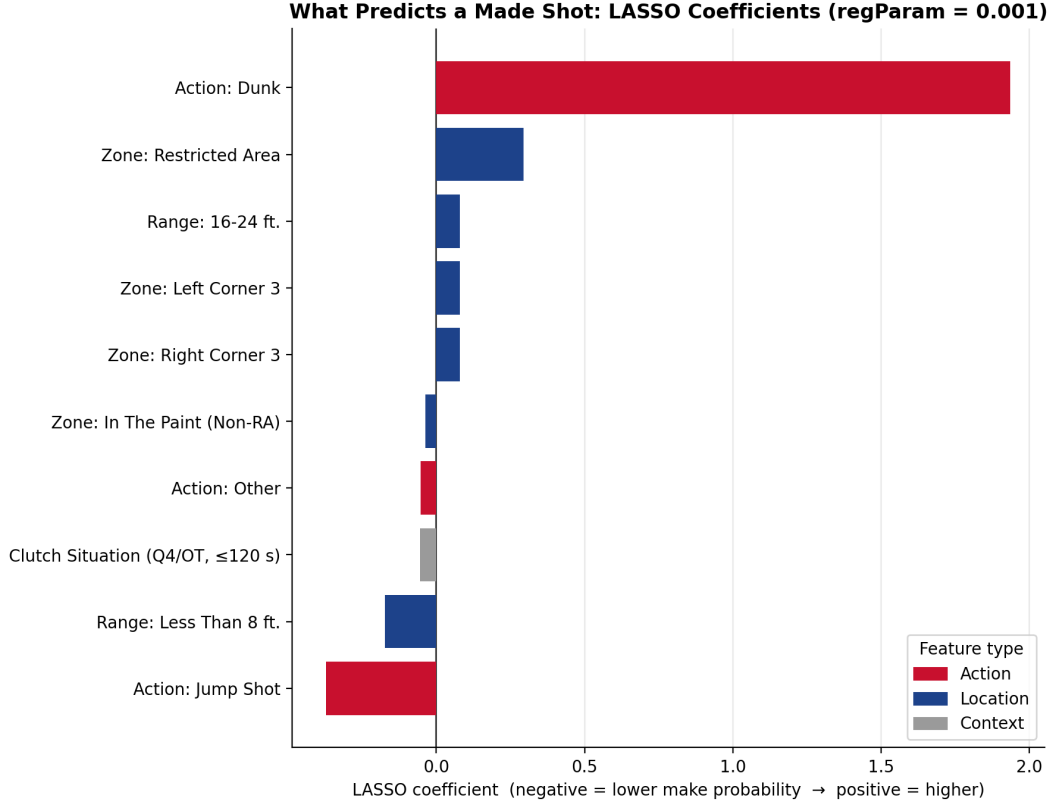


Figure 2: LASSO coefficients at $\text{regParam} = 0.001$, colored by feature category (Action, Location, Context). Positive values indicate higher make probability; negative values indicate lower. Features with $|\text{coefficient}| < 0.02$ are omitted as negligible for chart clarity, and the Back Court Shot coefficient (-1.34) is excluded as stated earlier being considered an outlier, unimportant feature.

Table 2: Binary classifier performance on the held-out 20% test split (full dataset, random seed 42). The always-guess-miss accuracy floor is approximately 54% (league-wide make rate 46%). Both models comfortably exceed this floor; their near-identical performance corroborates the result independently of model choice.

Model	ROC-AUC	F1	Accuracy
LogisticRegression	0.629	0.605	0.618
LinearSVC	0.62	0.601	0.611

Table 2 reports the classifier performance on the held-out test split. An ROC-AUC of 0.629 is modest but still an improvement from baseline 54% which would be a consistent miss guess (46% make rate). Without additional difficulty of shot via defender or contest data further improvement is unlikely. The agreement between the LogisticRegression and LinearSVC models point to a robust conclusion of feature importance in shot make prediction.

4.4 Player Archetypes: Who Thrived After the Revolution

Now I begin to look at player types based on different shots taken by that respective player. This is done over multiple seasons the player was in to see if the shot era inflection affected player type popularity in the league. Seperating the features via PCA and then running a K-means to group the players into 5 different archetypes.

Table 3: K-Means cluster shot profiles ($k = 5$) across 6,451 qualifying player-seasons (2004–2024). Zone shares are the fraction of a player-season’s shots taken from each zone; rows sum to approximately 1.00. Clusters are labeled based on their dominant zone concentration. See Figure 3 for the PCA scatter plot.

Archetype	N	Rim	Paint	Mid-Range	L-Corner 3	R-Corner 3	ATB 3
Modern Versatile Scorer	1,422	31.3%	19.1%	15.6%	3.8%	3.5%	26.6%
Rim-Dominant Big	1,136	59.3%	22.6%	14.6%	0.5%	0.5%	2.4%
Floor-Spacing Specialist	866	24.2%	8.6%	12.4%	11.6%	11.0%	32.3%
High-Volume Perimeter	1,581	21.3%	9.3%	25.1%	5.4%	4.9%	34.0%
Mid-Range Specialist	1,446	31.1%	14.8%	42.7%	1.5%	1.4%	8.6%

Table 3 shows the shot take percentages of all the players in the NBA from 2004-2024 creating five distinct archetypes. Each has a signature zone or role the first being the Rim-Dominant Big—a physical player getting most of their points close to the basket with near zero threes. A mid-range specialist shooting but not from three point range. The Floor-spacing specialist with a high three point volume from the corner. The high-volume perimeter creator usually the point guard shooting both three point and mid range shots. Finally the Modern Versatile scorer which has a balance of all the shot types. Respective shot percentages are shown in the table outting numbers to the descriptions.

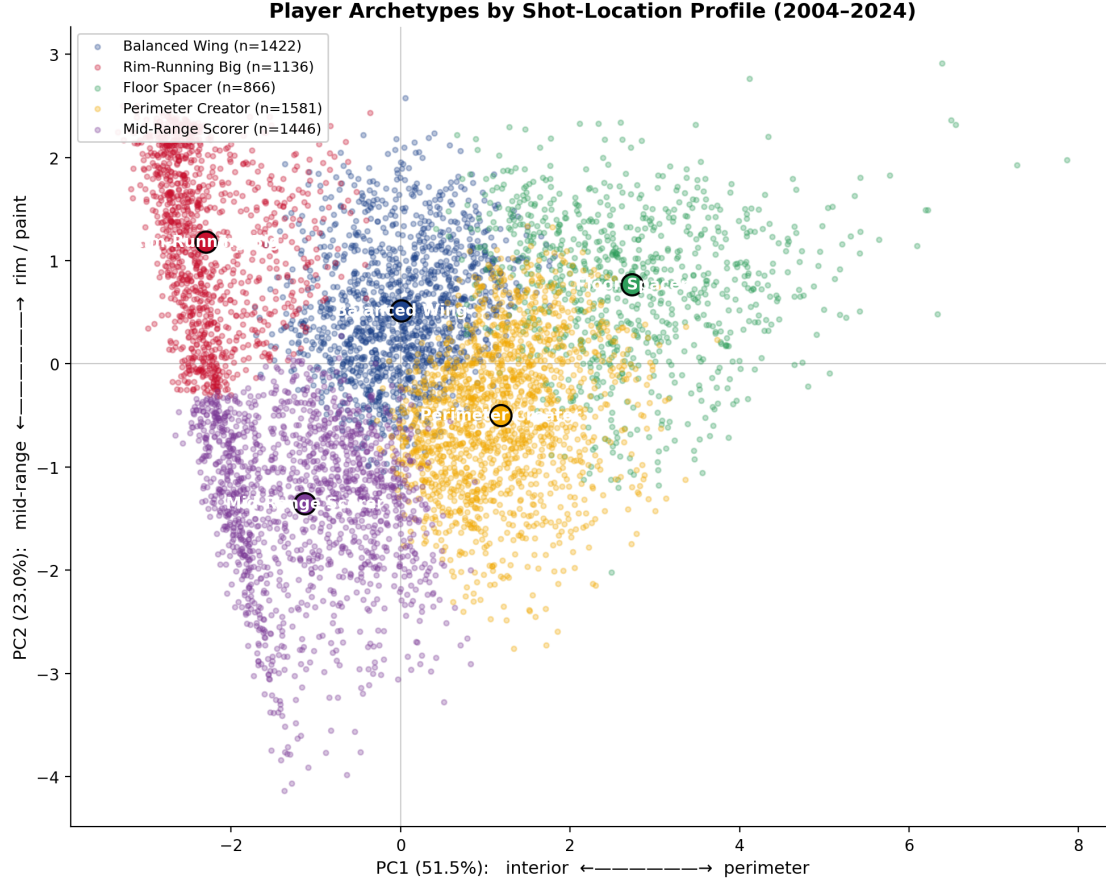


Figure 3: PCA scatter of 6,451 player-seasons colored by K-Means cluster ($k = 5$). PC1 and PC2 are the two dominant axes of shot-selection variation. Cluster separation reflects distinct shot profiles; see Table 3 for the zone-share interpretation of each archetype.

Figure 3 shows the graphical representation for the first 2 PCA vectors representing interior-perimeter shooting separation (PC1 51% of variability) on the x-axis and mid-range to rim separation (PC2 23% of variability) on the y-axis. Together holding 74.5% of the variation the geometry clearly shows the different player archetypes via color of the figure. There are overlaps and blurs at the border, however this is expected for a real life dataset like the NBA.

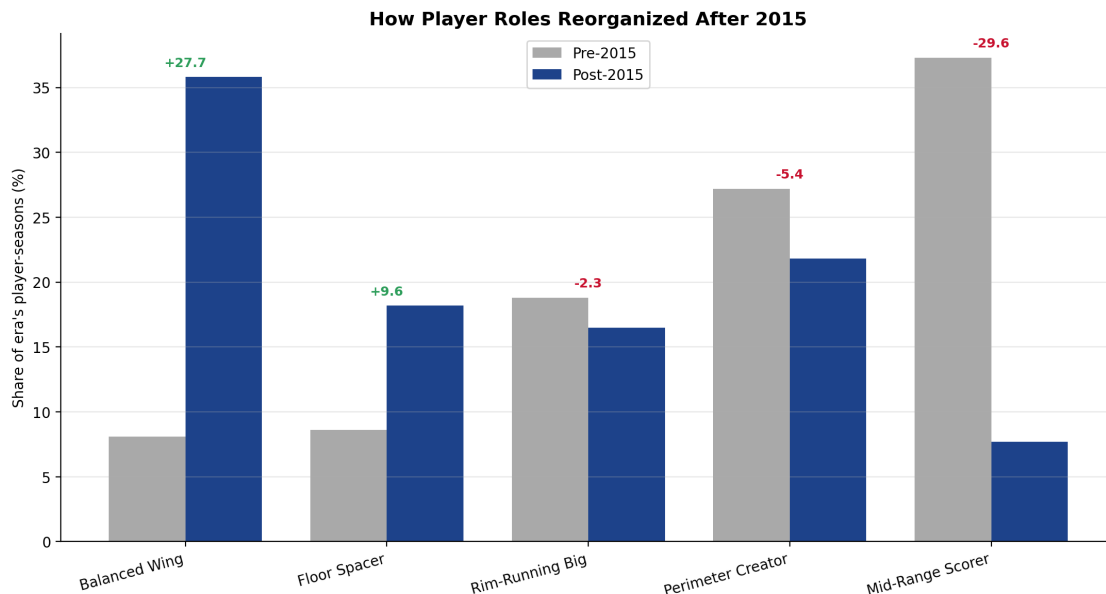


Figure 4: Archetype population shift before and after 2015. Each bar shows the percentage of qualifying player-seasons assigned to that archetype in the pre-2015 (seasons 2004–2014) and post-2015 (seasons 2015–2024) periods. The Mid-Range Specialist archetype declined 29.6 percentage points; the Modern Versatile Scorer grew 27.7 percentage points.

Figure 4 perfectly represents the change in player type before and after the 3pt transition. Mid-Range Scorer was the most populated archetype pre-era and dropped -29.6 points to almost non-existent current era numbers. High-volume perimeter and Rim-Dominant Big both see decreases as well however being much smaller at -5.2 and -2.3 respectively. The Floor-Spacing Specialist and Modern Versatile scorer saw increases in presence at +9.6 and +27.7 respectively showing the obvious increase in the balanced scorer. This reorganization of scoring types shows the power of transition the three point era had eliminating most players who could not adjust.

We must hold in account that the archetypes are built from shot location only, capturing no defensive presence resulting in an incomplete player profile. However, this should not affect the interpretation of the offensive change in the NBA requiring teams to also adjust defense. Changes in both should be expected and even possibly correlated by era.

5 Conclusion and Future Study

This study set out to answer three connected questions: whether the NBA’s shift toward the three-point shot was mathematically justified, when the league fully embraced it, and what that shift did to the types of players who define success. The evidence presented here points to an obvious answer: the transition was not only justified but effectively inevitable, it accelerated sharply in the mid-2010s, and it fundamentally reorganized player roles around the three-point line.

Taken together, the results form a single, coherent chain of cause, adoption, and consequence.

The underlying math always favored the three-point shot: mid-range attempts produced just 0.798 points per attempt compared to 1.054 for above-the-break threes establishing a structural efficiency gap built into the game itself. This advantage is reinforced by the shot-make model, where location and action type dominate contextual game features, indicating that where a shot is taken matters more than when or under what circumstances it occurs. The league’s behavior ultimately aligned with this reality, as the three-point attempt rate rose from 18.6% in 2004 to 39.4% in 2024, with the most dramatic acceleration occurring between 2015 and 2017 (Warriors dominance). Finally, this strategic shift reshaped the player landscape: the mid-range specialist archetype declined from 37.3% to 7.7% of player-seasons, while the modern balanced scorer expanded from 8.1% to 35.8%, reflecting a league now built around spacing and perimeter efficiency. These are not independent findings, they describe a single transformation from mathematical incentive to league-wide adoption to structural change in player roles.

Several limitations are important to acknowledge. First, the shot-make model’s modest predictive performance (ROC-AUC = 0.629) reflects a key data constraint: the absence of defender proximity and contest information, meaning the model captures the value of shot location and type, not true shot difficulty. Second, while the alignment between efficiency math and league behavior is clear, this study shows association rather than proving a direct causal relationship for the change in strategy. Third, cross-era comparisons are influenced by external factors, including rule changes such as the 2004–05 hand-check adjustment and shortened seasons in 2012 and 2020 which were briefly mentioned. Finally, the player archetype analysis is based solely on shot location and relies on a fixed 2015 cutoff, which simplifies a more gradual and multidimensional evolution of player roles.

Future work can build directly on these gaps. The most important extension would be incorporating defender tracking and contest data to model true shot quality, allowing for a clearer distinction between shot selection and shot difficulty. A second direction is to examine how ball movement and team structure evolved alongside shot selection during this transition. Finally, a more detailed robustness and fairness analysis across eras, zones, and player groups could test the stability of these findings and identify where the model performs differently. Taken together, these extensions would move the analysis from describing the three-point revolution to fully explaining the mechanisms behind it, offering a deeper understanding of how strategy evolves in professional basketball.

6 References

Samangy, D. (2025). *NBA_Shots_04_25* [Data set]. GitHub. https://github.com/DomSamangy/NBA_Shots_04_25

National Basketball Association. (2025). *NBA Stats API*. <https://www.stats.nba.com>

Appendix: Representative Code

The following code chunks are representative PySpark segments from the analysis scripts committed to the project repository. Full scripts are available in the repository at `02_processing/process_shots.py`, `03_shot_model/shot_model.py`, `03_shot_model/feature_engineering.py`, and `05_clustering/archetypes.py`.

A.1 Zone Efficiency Aggregation (`02_processing/process_shots.py`)

```
# Aggregate zone efficiency: FG% and points per attempt by BASIC_ZONE
zone_stats = (
    shots
    .groupBy("BASIC_ZONE")
    .agg(
        F.count("*").alias("attempts"),
        F.round(F.avg("SHOT_MADE_INT") * 100, 1).alias("fg_pct"),
        F.round(F.avg("EXPECTED_PTS"), 3).alias("pts_per_attempt"),
    )
    .orderBy(F.desc("pts_per_attempt"))
)
zone_stats.write.csv(TABLES_PATH + "fg_pct_by_zone", header=True, mode="overwrite")
```

A.2 LASSO Regularization Path (`03_shot_model/shot_model.py`)

```
# Sweep regParam values and record the coefficient vector at each penalty
REG_PATH = [0.05, 0.01, 0.005, 0.001, 0.0005, 0.0001]
HEADLINE_REG = 0.001

path_rows = []
for reg in REG_PATH:
    lasso = LogisticRegression(
        featuresCol="features", labelCol=LABEL_COL,
        elasticNetParam=1.0, # pure L1 = LASSO
        regParam=reg, maxIter=100,
    )
    pipeline = Pipeline(stages=build_feature_stages() + [lasso])
    model = pipeline.fit(train)
    lr_model = model.stages[-1]
    coeffs = lr_model.coefficients.toArray()
    feat_names = extract_feature_names(model.transform(train.limit(1)))
```

```

    for name, coef in zip(featur_names, coeffs):
        path_rows.append((reg, name, float(coef)))

path_df = spark.createDataFrame(path_rows, ["regParam", "feature", "coefficient"])
path_df.write.csv(LASSO_PATH_OUT, header=True, mode="overwrite")

```

A.3 PCA and K-Means Archetype Pipeline (05_clustering/archetypes.py)

```

# StandardScaler → PCA (5 components) → K-Means (k=5, seed=42)
N_PCA = 5
CHOSEN_K = 5
SEED = 42

scaler = StandardScaler(inputCol="features", outputCol="scaled",
                        withMean=True, withStd=True)
players = scaler.fit(players).transform(players)

pca_model = PCA(k=N_PCA, inputCol="scaled", outputCol="pca_features").fit(players)
players = pca_model.transform(players).cache()

# Report explained variance per component
ev = pca_model.explainedVariance.toArray()
for j, frac in enumerate(ev):
    print(f" PC{j+1}: {frac:.1%} (cumulative {ev[:j+1].sum():.1%})")

# Final clustering
km_model = KMeans(featuresCol="pca_features", predictionCol="cluster",
                  k=CHOSEN_K, seed=SEED).fit(players)
assigned = km_model.transform(players)

# Era comparison: pre-2015 vs post-2015 cluster population shares
era_totals = {r["ERA"]: r["count"]}
for r in assigned.groupBy("ERA").count().collect():
    era_counts = (
        assigned
        .groupBy("cluster", "ERA")
        .count()
        .withColumn("pct", F.round(F.col("count") / era_totals[F.col("ERA")] * 100, 1))
    )

```